



Manila City House Prices: A Machine Learning Analysis of the Current Market Value for Improvements

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Abstract. The goal of the study is to develop an intelligent system that employs machine learning techniques to predict house prices in Manila City. The dataset contains attributes such as floor area, no. of storeys, classification, type of structure, additional floors or areas, and tag ID. Machine learning algorithms such as Linear Regression, Bayesian Ridge, Gradient Boost, and Lasso Regression were utilized to construct the predictive model. Feature selection methods and Genetic Algorithm optimization were also implemented. The model's performance was evaluated using different performance metrics, and the findings indicated that Gradient Boost exhibited the most optimal performance with an optimized R-squared value of 0.7508, and EVS value of 0.7640. Based on the evaluation results, the proposed model with genetic algorithm optimization can provide accurate predictions for the prices of houses in Manila City, and could be a useful tool for both buyers and sellers in the real estate industry.

Keywords: Real state appraisal · Current market value · Regression models · Manila City house prices

1 Introduction

Property investment is a major financial decision that requires a significant amount of initial capital [1]. Thus, any decision to invest in property should be based on a careful assessment, however, accurately determining the market value of these improvements can be a complex and time-intensive task, often involving factors such as residential, structural and environmental [2].

These various factors are commonly examined by traditional real estate appraisal methods, however, such manual procedures may be susceptible to the biases and interests of appraisers, posing a potential risk for a biased or subjective evaluation of a property. This may lead to losses for investors or households involved in the transaction [3]. The advancement of AI technology increases the likelihood of a transition in labor demand from manual and repetitive duties to digital tasks that necessitate skill sets like data

analysis and algorithm development [4]. Therefore, developing practical algorithms and automated models that can provide objective and impartial real estate appraisals has become increasingly important, not only for improving the accuracy of valuations but also for reducing the burden of labor-intensive work [5].

In this research, we aim to predict house prices in Manila City, Philippines using some of the machine learning techniques. Our analysis will focus on identifying key features that influence house prices in Manila City, and using this information to develop a model that can predict prices for properties in the city. This research will provide valuable insights for real estate investors, property developers, and academic researchers interested in the housing market in Manila City. Additionally, we will also explore the use case of this research on government's urban planning [6].

2 Review of Related Literature

Recent advancements in machine learning models and the emergence of big data have provided new opportunities for real estate industry [7]. Research studies utilizing statistical learning approaches to analyze the housing market have been conducted in various locations, such as the United States [1, 2, 5–9], Europe [3, 10], Australia [11, 12], and China [13]. Nonetheless, studies applying data analytics and machine learning models to examine the housing market in the Philippines are not widely available in the literature.

Several studies have been conducted to investigate the variables that contribute to housing prices. For example, Zhang conducted a multiple linear regression of the variables that influences the value of residential properties. The findings revealed that property size, location, no. of floors, no. of rooms, and availability of facilities were key determinants of the property's appraisal value [2].

Satish developed a model for predicting housing costs by utilizing various machine learning algorithms, including Gradient Boost, XGBoost, Lasso regression, and neural networks. The results of the analysis showed that the Lasso regression algorithm consistently outperformed the other models in terms of accuracy for predicting housing costs [9]. Fan et al. analyzed the different machine learning algorithms such as Lasso Regression, Ridge Regression, and XGBoost with the use of optimal parameters [6].

Another study by Garcia compared the performance between linear regression and ensemble models for mass-appraisal algorithm techniques. According to its findings, machine learning algorithms are more suitable for dealing with the nonlinearities present in complex real estate market data than traditional linear models. Bagging-based algorithms, such as random forest and extra-trees regressor, exhibit overfitting issues. In contrast, boosting-based algorithms demonstrate superior performance and exhibit lower levels of overfitting [10].

These studies provide valuable insights into the factors that influence housing prices in Manila City and highlight the capability of machine learning techniques in predicting the market value of house improvements. However, there is a need for further research to explore the specific factors that contribute to the market value of house improvements in Manila City and to develop accurate models for predicting these values. The objective of this study is to address the gap in existing literature by employing machine learning algorithms to the current market value of house improvements in Manila City. In particular,

we will contrast our proposed methodology with the machine learning models utilized in prior studies. Our approach includes feature reduction techniques to determine the essential features that impact house prices in Manila City, as well as genetic algorithm optimization to identify key parameters in a predictive model.

Overall, this review demonstrates the importance of understanding the factors that contribute to the market value of housing improvements and highlights the potential of machine learning techniques in predicting these values.

3 Methodology

The proposed methodology for predicting house prices in Manila City involves a multi-step process, shown below, done using Python, including the pandas, numpy, and scikit-learn libraries. Jupyter notebook was used as the development environment for the analysis (Fig. 1).

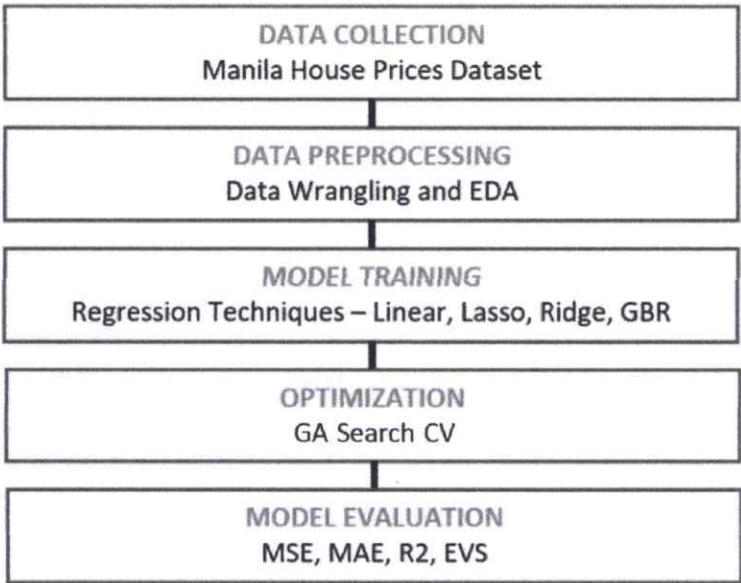


Fig. 1. Research methodology framework

3.1 Data Collection

The dataset used in this study is from the Department of Public Works and Highways Public and Private Partnership Service (DPWH-PPPS) and it pertains to the current market value of the improvements affected by the NLEX-SLEX Connector Road Project right-of-way.

The location of the properties in the dataset is situated in Manila City and it includes various features such as floor area in sqm, number of storeys, classification, type of structure, additional floors or areas, and tag ID. The initial dataset prior to data preprocessing consists of 1,331 improvements (Table 1).

Table 1. List of features

Feature name	Data type	Description
Tag ID	object	A unique identification number assigned to each property in the dataset for tracking and referencing purposes
Floor area	float64	Refers to the total floor area of the house, usually measured in square meters
No. of storeys	int64	Refers to the number of levels in the house, including the ground floor
Additional area	object	Refers to any additional floors or areas in the house, such as a mezzanine, attic, roof deck or roof slab
Classification	object	Refers to the classification of the house, such as residential or commercial
Type of structure	object	Refers to the type of construction material used in building the house, such as concrete, semi-concrete, or wood

3.2 Data Preprocessing

Preprocessed the data by cleaning, deleting missing values, and transforming variables as needed. This step ensures that the dataset is ready for analysis and model building. It should be noted that there are numerous instances of missing values in the target variable - the price of the properties. Although it is possible to replace these missing values with imputed values, this approach could introduce additional bias in the input data [12]. Therefore, any observations with missing values in the ‘Estimated Cost’ column are eliminated from the dataset to enhance the reliability of the analysis. The final dataset used for the model was narrowed down into 1,067 improvements.

Furthermore, the Tag ID feature, which serves as a unique identifier for each house in the dataset, is not included in the regression analysis. The Tag ID feature is used exclusively for data management purposes and is therefore not pertinent to the housing market analysis. As a result, it is removed from the feature selection process and the subsequent regression analysis.

Categorical data are typically represented as text values in a dataset. For Manila City House Prices dataset, features such as additional area, classification and type of structures are needed to be converted in order to use such data in machine learning models. To transform categorical data into numerical values, two pre-processing techniques were utilized in this study: Label Encoder and One-Hot Encoder. ColumnTransformer allowed us to use pre-processing technique to different columns of a dataset while avoiding the dummy variable trap.

One-Hot Encoding produced seven columns for the additional area feature, two columns for classification, and three columns for type of structure. In order to avoid the dummy variable trap, one column for each feature has been removed. The final dataset has a shape of (1067, 11).

Feature reduction is another important step in data preprocessing that involves the selection of a subset of the most significant features from a larger feature set in a dataset.

The aim is to simplify the dataset and remove irrelevant or redundant features that may not contribute to the predictive power of a machine learning model.

The feature reduction technique used was Backward elimination. It involves iteratively removing features from a model until all the remaining features satisfy a predetermined significance level, set at 0.05. The elimination process starts with a full model that includes all features, and the significance of each feature is assessed using a statistical test such as the p-value. Features that do not meet the significance level are removed, and the model is re-fitted with the remaining significant features. This iterative process continues until all the features that remain in the model are statistically significant, and no further features can be removed without compromising the model's performance. By using this approach, eleven features generated from one hot encoder has been reduced into six statistically significant features; linear constant (column 0), with Mezzanine and Roof Slab from Additional Floor feature (column 4), Residential from classification feature (column 7), Semi-concrete from type of structure feature (column 8), floor area (column 10) and number of storeys (column 11).

3.3 Model Training

The performance of four distinct machine learning techniques for predicting the target variable has been evaluated: Linear Regression [2], Lasso Regression [9], Gradient Boosting Regression [10], and Bayesian Ridge Regression [6].

To ensure a fair comparison of the algorithms, we used both Hold-out validation and K-fold cross-validation with $K = 10$ [14]. In each fold, the data was randomly partitioned; training data consisted of 80% of the dataset, while the remaining 20% was assigned to the testing data [15]. The process was repeated ten times, with each fold serving as the test set once.

3.4 Optimization

Genetic algorithm is a type of optimization technique inspired by the natural selection process [16]. The algorithm starts with a population generated by randomly selecting genes from a uniform distribution. An individual is the term used for each solution, and a generation is the term used for the entire set of individuals. The individuals are evaluated based on a fitness function, which measures how well they perform on the given task. In the context of hyperparameter tuning, the fitness function is typically a measure of model performance such as accuracy, precision, recall, or R^2 score. The genetic algorithm then evolves the population through several generations, using three main operators: selection, crossover, and mutation. Selection involves selecting the fittest individuals from the population for the purpose of creating a new generation. Crossover is the process of joining the genetic material of two individuals at random to form a new individual. Mutation is the process of modifying an individual's genetic material at random in order to create a new one [17].

The GASearchCV class implements a genetic algorithm search for hyperparameter tuning in a scikit-learn pipeline. The estimator parameter is the estimator to tune, and the param_grid parameter is a dictionary of hyperparameter ranges to search over. The other

parameters control the genetic algorithm search, including the number of generations, the population size, and the fitness function or scoring [18].

GASearchCV works by creating a random initial population of hyperparameters and then evolving that population over several generations using the three genetic operators. At the end of the search, the best set of hyperparameters is returned. The hyperparameters for each algorithm and the corresponding ranges for the search were as follows:

- Multiple Linear Regression: ‘normalize’, True or False
- Bayesian Ridge Regression: alpha_1 and alpha_2, ranges from 1e-6 to 1e-2, lambda_1 and lambda_2, ranges from 1e-6 to 1e-2
- Gradient Boosting Regression: n_estimators, learning_rate, max_depth, and subsample, with values from 50 to 200, 0.01 to 0.1, 3 to 7, and 0.5 to 1, respectively.
- Lasso Regression: alpha, ranges from 1e-6 to 1e-2.

The performance of each algorithm was assessed by computing the R-squared values for each fold and then reporting the resulting average and standard deviation.

3.5 Model Evaluation

To assess the effectiveness of our regression models, we used the following metrics:

- The MAE or mean absolute error, measures the average magnitude of our predictions’ errors and is calculated by averaging the absolute difference between the predicted and true values. Lower MAE indicates a higher performance [19].
- The MSE or mean squared error, calculates the average squared difference between predicted and true values by taking the mean of the squared differences. Lower MSE indicates improved performance.
- The R^2 or R-squared, is a measure of how well the model fits the data, and is computed as the ratio of the explained variance to the total variance. The scale is between 0 and 1, where higher values indicate excellent performance.
- The EVS or Explained variance score, explains the error dispersion in a particular dataset. The formula is 1 minus the ratio of the residual variance to the total variance. The scale is between 0 and 1, where higher values indicate excellent performance.

By using all of these metrics, we aimed to get a comprehensive understanding of how well our models were performing, and to identify which model was the best for Manila city house prices task.

4 Results and Discussion

Table 2 and 3 present the results of our model evaluation using the MSE, MAE, R-squared, and EVS metrics for each of the four regression models that we used (Linear Regression, Bayesian Ridge, Gradient Boost, and Lasso Regression) using default parameters. Tables 2 and 3 were evaluated through Cross-fold validation and Hold-out Validation, respectively [14].

For Cross-fold validation, we can see that the Bayesian Ridge model performed the best overall, achieving the highest R-squared and EVS scores, while the Linear regression

model and Gradient Boost presented a lowest MSE and MAE, respectively. However, all four models performed relatively well, with R^2 scores ranging from 0.6625 to 0.7378, and EVS scores ranging from 0.6689 to 0.7378.

Table 2. Cross-fold validation evaluation performance through default parameters

Model	MAE	MSE	R^2	EVS
Linear regression	417966.71	1104813484865.54	0.7333	0.7373
Bayesian ridge	418124.25	1104850876069.80	0.7338	0.7378
Gradient boost	400171.3	1251050671665.5	0.6625	0.6689
Lasso regression	417966.82	1104814155048.98	0.7333	0.7373

For Hold-out validation, we can see that the Gradient Boost model performed the best overall, achieving the lowest MSE, MAE, and highest R-squared and EVS scores.

Table 3. Hold-out validation evaluation performance through default parameters

Model	MAE	MSE	R^2	EVS
Linear regression	450278.72	1392541484222.29	0.7898	0.7925
Bayesian ridge	450764.98	1405205956080.20	0.7878	0.7906
Gradient boost	419569.27	1096599623150.95	0.8344	0.8356
Lasso regression	450278.87	1392557148349.98	0.7898	0.7925

With the use of GASearchCV for optimization, the best hyperparameters for each algorithm that generated highest value of R-squared are:

- Multiple Linear Regression: ‘normalize’: False
- Bayesian Ridge Regression: ‘alpha_1’: 0.0004915314720013189, ‘alpha_2’: 0.0021894794843560963, ‘lambda_1’:0.005960462584099383, ‘lambda_2’: 0.009450054454871687
- Gradient Boosting Regression: ‘learning_rate’: 0.033298366669962574, ‘subsample’: 0.5634666800882826, ‘n_estimators’: 57, ‘max_depth’: 4
- Lasso Regression: alpha of 0.0099939746971181

Tables 4 and 5 present the results of our model evaluation using the MSE, MAE, R-squared, and EVS metrics for each of the four regression models that we used (Linear Regression, Bayesian Ridge, Gradient Boost, and Lasso Regression) using optimized parameters from GASearchCV. Tables 4 and 5 were evaluated through Cross-fold validation and Hold-out Validation, respectively [14].

For Cross-fold validation, we can see that the Gradient Boost model performed the best overall, achieving the highest R-squared and EVS scores and lowest MSE Score, while the Linear regression model presented a lowest MAE. However, all four models

performed relatively well, with R^2 scores ranging from 0.7333 to 0.7508, and EVS scores ranging from 0.7373 to 0.7640.

Table 4. Cross-fold validation evaluation performance through optimized parameters

Model	MAE	MSE	R^2	EVS
Linear Regression	417966.71	1104813484865.54	0.7333	0.7373
Bayesian Ridge	418124.64	1104851889106.48	0.7338	0.7378
Gradient Boost	429877.67	1093145177225.57	0.7508	0.7640
Lasso Regression	417966.71	1104813491630.05	0.7333	0.7373

For Hold-out validation, we can see that the Linear Regression model performed the best overall, achieving the lowest MSE, MAE, and highest R-squared and EVS scores.

Table 5. Hold-out validation evaluation performance through optimized parameters

Model	MAE	MSE	R^2	EVS
Linear regression	450278.72	1392541484222.29	0.7898	0.7925
Bayesian ridge	450766.49	1405245581297.19	0.7878	0.7906
Gradient boost	503734.71	1528845463101.02	0.7692	0.7726
Lasso regression	450278.72	1392541640769.94	0.7898	0.7925

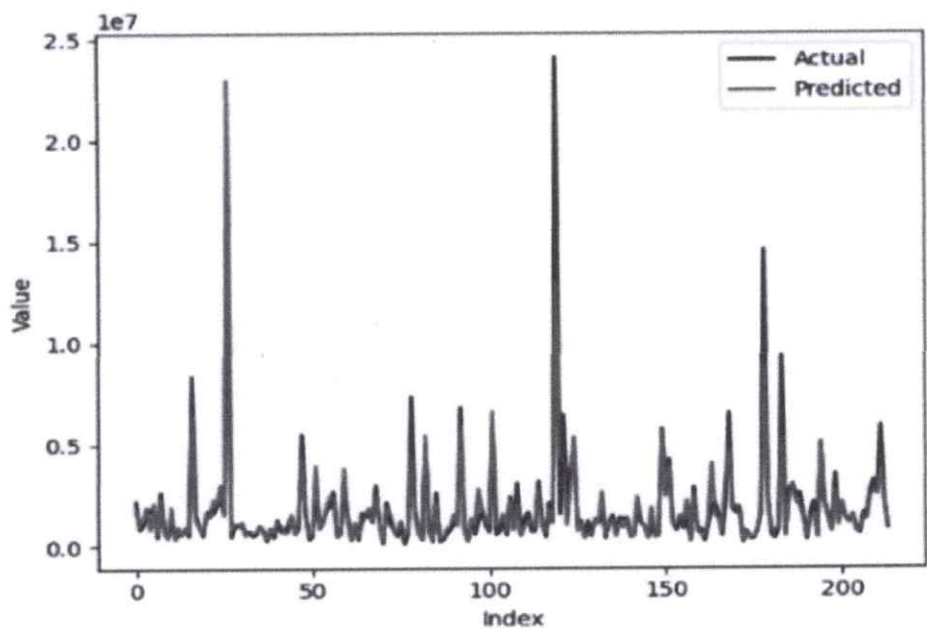


Fig. 2. Trend lines of actual vs. predicted value for optimized gradient boost model

As the optimized gradient boost model performed best overall using Cross-fold Validation, we can also see in Fig. 2 that it generated prediction results that are generally

consistent with the trend of real value since the trend moves along the same direction with each other.

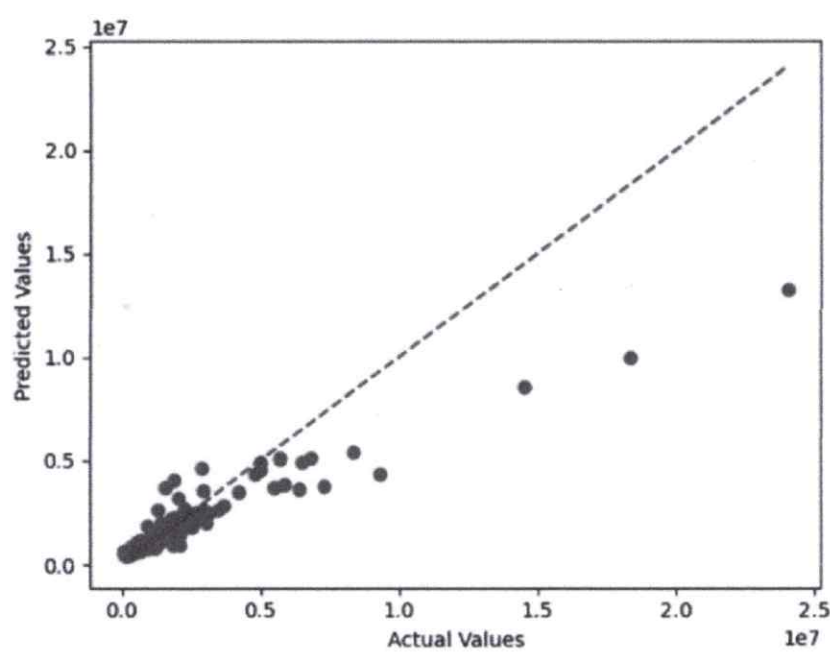


Fig. 3. Scatter plot of actual vs. predicted value for optimized gradient boost model

We can also see that it generated a scatter plot Fig. 3, containing relatively accurate predictions. The plot is closer to the diagonal line, which suggests that the model’s predictive performance is good.

5 Conclusion

In this study, we employed machine learning models to develop an intelligent system that predicts house prices in Manila City. The dataset used in this study was obtained from the Department of Public Works and Highways Public-Private Partnership Service (DPWH-PPPS) and it pertains to the current market value of the improvements affected by the NLEX-SLEX Connector Road Project right-of-way. It includes various features such as floor area in sqm, number of storeys, classification, type of structure, additional floors or areas, and tag ID. After performing feature engineering and Genetic algorithm optimization, it was determined that Gradient Boosting Regressor was the top-performing model in predicting house prices in Manila City, with an R-squared score of 0.7508 and EVS value of 0.7640. This study demonstrated the usefulness of machine learning in predicting house prices and could aid in making informed decisions for real estate investment in Manila City. In future studies, enhancing the performance of the model can be achieved by integrating a more comprehensive dataset and diverse features.

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